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Perspective

World models for biomedicine

Ayush Noori,[1,2,9] Nic Fishman,[1,3,9] Ada Fang,[1,4,5,10] Lukas Fesser,[1,5,6,10] and Marinka Zitnik[1,5,7,8,*]
[1]Department of Biomedical Informatics, Harvard Medical School, Boston, MA, USA
[2]Department of Engineering Science, University of Oxford, Oxford, UK
[3]Department of Statistics, Harvard University, Cambridge, MA, USA
[4]Department of Chemistry and Chemical Biology, Harvard University, Cambridge, MA, USA
[5]Kempner Institute for the Study of Natural and Artificial Intelligence at Harvard University, Allston, MA, USA
[6]Applied Mathematics, Harvard John A. Paulson School of Engineering and Applied Sciences, Allston, MA, USA
[7]Broad Institute of MIT and Harvard, Cambridge, MA, USA
[8]Harvard Data Science Initiative, Cambridge, MA, USA
[9]These authors contributed equally
[10]These authors contributed equally
*Correspondence: marinka@hms.harvard.edu
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SUMMARY

Biological systems respond dynamically to changes in their environment, and predicting these responses before complex interventions are applied could change how computational models are used in biology and medicine. In artificial intelligence, world models represent the state of a system and simulate how it evolves under alternative actions. We envision extending this paradigm to biomedicine, with world models that span molecular, cellular, tissue, and clinical scales and simulate biological trajectories under alternative interventions. Building such models will require methods that combine observational data with interventional and longitudinal measurements across modalities. Biomedical world models could open new modes of inquiry by predicting how biological systems respond to novel perturbations, searching for interventions that drive them toward desired states, and planning experimental campaigns over extended time horizons.

INTRODUCTION

Since the Scientific Revolution of the 16th and 17th centuries, experimentation, or systematic intervention followed by observation, has enabled scientists to advance human knowledge of the natural world. Scientists have built experimental platforms that perturb genes, administer drugs, alter environments, or modify behavior, and that measure the resulting consequences across molecules, cells, organisms, and patients. As Claude Bernard wrote in the founding text of experimental medicine, the sciences "all try to reach knowledge of the law of phenomena so as to foresee, vary, or master phenomena."[1,2]

Learning such laws that govern how biological systems respond to intervention requires exploring an experimental space that is often far larger than can be sampled directly. Practical, technical, and ethical constraints limit how much of this space can be tested experimentally. As interventions vary by combination, dose, timing, and biological context, the number of possible experiments grows combinatorially. Drugs can be administered at different doses, developmental stages, or disease states; genes can be perturbed individually or together; and the same experiment can produce different results across cell types, tissues, and patients. Experimentally measured biomedical knowledge therefore covers only a sparse and uneven sample of all biological states. For example, the Human Reference Interactome is estimated to only represent between 2% and 11% of the binary protein interactome[3]; only 0.1% of the 71 million possible variants in the human genome are annotated in ClinVar[4]; the largest combinatorial perturbation screens still test only 2% of the nominal pairwise protein-coding gene space[5]; and all licensed drugs act through only 652 genes, less than 15% of the estimated druggable genome and only 3.2% of all protein-coding genes.[6] Sampling is also uneven for reasons unrelated to measurement. Historical, social, and economic incentives have directed most research toward 10% of the human genome,[7] leaving at least one in three human genes functionally poorly understood.[8]

Navigating this space requires models that can reason beyond experiments already performed. Biomedical world models, or artificial intelligence (AI) systems that simulate how biological systems evolve under interventions, offer this capability (Figure 1). Rather than approximating specific biological functions, world models represent how systems change under varying interventions and generate trajectories tracing out possible futures. In robotics and autonomous driving, world models can plan over long time horizons and generalize across diverse environments by learning compact latent dynamics that enable this kind of simulation (Box 1).[9] A biomedical analog would simulate the consequences of a proposed sequence of actions, such as a set of gene knockouts, a drug combination, or a change in experimental conditions. Building such models is difficult. While probabilistic models of images, natural language, and protein structures can be learned from observational datasets, world models are

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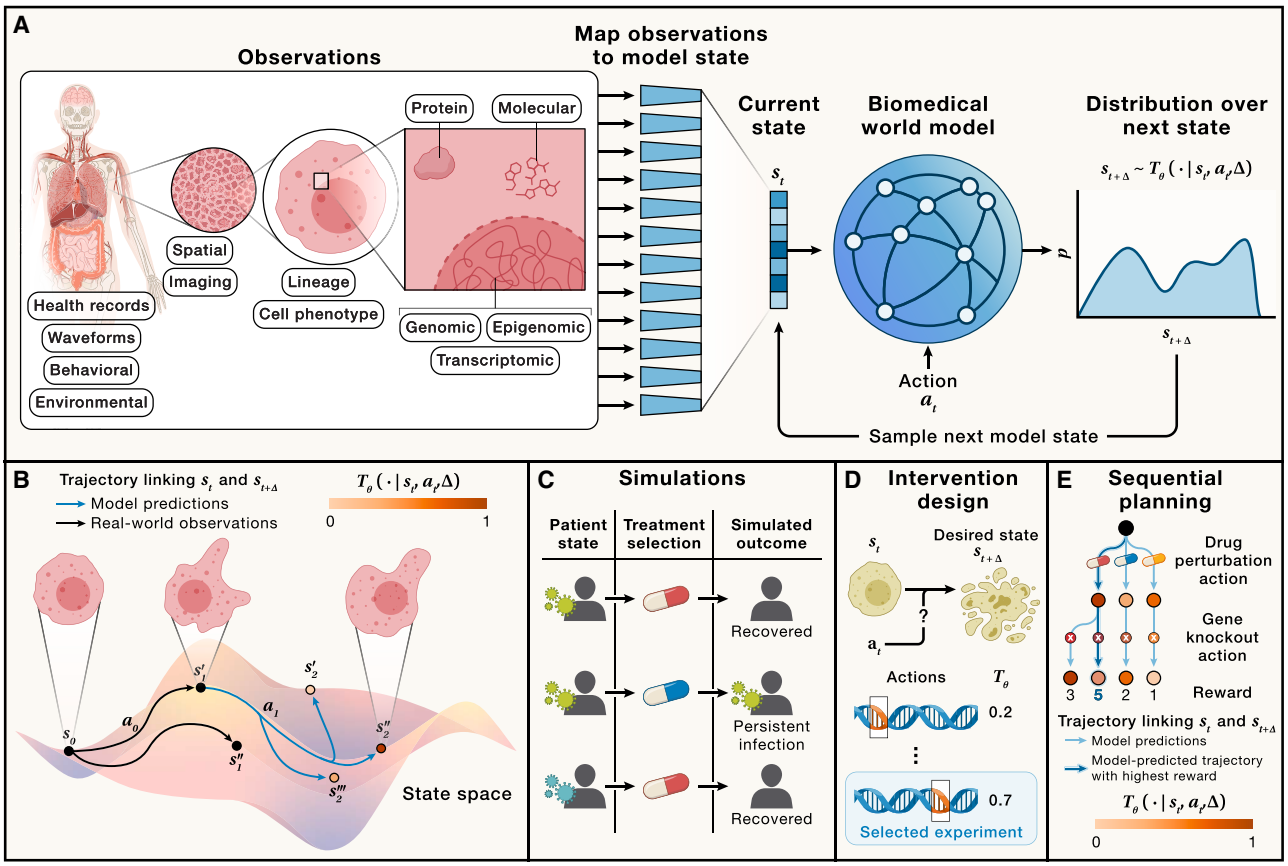


Figure 1. Biomedical world models simulate action-conditioned biological trajectories

(A) A biomedical world model integrates observations across modalities to construct a representation of the current biological state, s_t. Given this state and an action a_t, the model predicts a distribution over possible next states, $s_{t+\Delta} \sim T_\theta(\cdot|s_t, a_t, \Delta)$.

(B) Repeated application of action-conditioned transitions generates trajectories through the state space of a cellular world model.

(C) In simulation, alternative actions can be applied to the same initial state, and the same action can be evaluated across different initial states to compare predicted outcomes.

(D) In intervention design, the model can be searched in reverse: given a current state and a desired future state, candidate actions are evaluated according to their predicted ability to induce that state.

(E) In sequential planning, predicted states are used as starting points for future simulation, allowing estimation of long-horizon evolution.

conventionally trained on longitudinal sequences of interleaved interventions and observations across diverse contexts.

Biomedical world models would not remove the need for experiments, but they could help human scientists and AI systems, including agentic AI scientists,[14] accelerate discovery. AI scientists are agents that carry out research processes autonomously; current systems synthesize literature,[15] generate hypotheses and analyze data,[16–18] and can even connect to automated laboratories.[19–22] While AI scientists can propose thousands of hypotheses in a single run, the challenge is deciding which experiments merit physical execution given the practical limits of wet lab experimentation.[23,24] World models could provide a foundation for this kind of prioritization, allowing biological systems to be explored over extended time horizons and the consequences of different experiments to be simulated and compared before the most promising are performed.

In this perspective, we first identify the defining properties of world models and distinguish them from related classes of models, including mechanistic simulations, foundation models, and digital twins. Under this definition, virtual cell, virtual embryo, and virtual patient approaches are aspirational world models, and we use them as case studies to examine the data, model architectures, and evaluation frameworks required to build world models. Three challenges emerge as distinct and essential to biomedical world modeling. Biological state is often only partially observed through one or a handful of assays. Many biological measurements destroy the sample, so data are typically cross-sectional rather than longitudinal. Current biomedical data are either observational, providing rich coverage of states without interventions, or interventional, covering comparatively few states. From a data generation perspective, we discuss new measurement technologies that could close these gaps at scale. From a computational perspective, we describe how multimodal data integration and models that learn jointly from observational and interventional data offer routes forward. If these challenges in observing richer states, generating longitudinal measurements,

Box 1. World models beyond biology

The term "world model" is used across several research domains, but what constitutes the "world" and the capabilities expected of the model vary substantially.

Video games: In video games, reinforcement learning agents learn by trial and error: they observe images, issue control inputs, and receive scores. A world model encodes each observation into a latent state and predicts future states and rewards under candidate actions, so behaviors can be internally tested in the model rather than in the environment itself. Ha and Schmidhuber rejuvenated interest in world models by showing that a game controller could be trained from scratch inside a world model.[10] Dreamer extends this idea by coupling a learned dynamics model with an actor that selects actions and a critic that scores predicted futures, training both on model-generated trajectories.[9] In Minecraft, where collecting diamonds requires a long chain of actions with intermediate barriers, Dreamer was the first algorithm to succeed from scratch, without human data or curricula. Games provide an unusually favorable setting for world models because evaluation is cheap and rapid: learned behaviors can simply be deployed in the game and scored.

Robotics: Manipulating objects with a robotic arm poses a related problem of physical control: robots observe their workspaces through cameras and act through motor commands. Unlike in video games, data collection requires operation of the physical robot, making large-scale trial and error costly. Rather than exclusively trying to scale robot interaction data, recent systems pre-train on abundant video of the world and specialize using much smaller amounts of robotic data. Current models take different approaches in what they predict. Cosmos predicts future video: pre-trained on 20 million h of video and adapted to a particular robot, it can serve as a visual simulator of what the robot's camera would observe under different actions.[11] V-JEPA 2 instead predicts future latent states rather than pixels.[12] Adapted with 62 h of robot recordings, it can be used directly for control by selecting the action sequence whose predictive state most closely matches the state of a goal image, replanning after each step.

Autonomous driving: In autonomous driving, the vehicle collects observations of the road through cameras and acts by steering, accelerating, and braking while interacting with other drivers whose behavior it does not control. Ordinary driving can be recorded at scale because fleets can passively collect demonstrations from human drivers, but this means crashes and other dangerous situations are rare or entirely absent in the data. GAIA-1, a generative world model trained on 4,700 h of driving data, predicts future observations conditioned on the current scene and a candidate maneuver or text prompt.[13] It can generate counterfactual situations that are scarcely represented in the training data: forced to steer into oncoming traffic, for example, the model generates the approaching vehicle swerving to avoid collision. Such models can therefore serve as neural simulators for generating synthetic, including adversarial, scenarios to train and evaluate driving systems. Autonomous driving is distinctive because the most critical scenarios are too dangerous, rare, or expensive to reproduce in the real world.

and expanding interventional coverage can be overcome, world models could become an important tool for exploring biological interventions.

DEFINING A BIOMEDICAL WORLD MODEL

The term *world model* has a long history in the psychological and computational sciences. Its origins are usually traced to psychologist Kenneth Craik,[25] who proposed that an organism possessing a *small-scale model* of external reality could internally test actions before executing them. The idea entered AI through two related traditions: means-ends analysis, which began as a psychology of human problem solving,[26] and the effort to equip machines with explicit representations of their environment.[27,28] The term itself crystallized in robot planning in the 1970s, where systems searched over spaces of symbolic world models to identify sequences of actions to achieve a goal.[29] Subsequent pioneering work in reinforcement learning developed adaptive networks capable of learning an internal model of the world.[30–34] In the early 1990s, some formulations distinguished the world model, referring to an agent's internal representation of its environment, from an *action model* describing how that representation changed under actions.[32,35] This distinction faded as the field increasingly organized around *model-based* and *model-free* learning.[36] The modern resurgence of world models in computer science is often linked to Ha and Schmidhuber,[10] who learned a compact representation of an environment and a dynamics model within that representation space, and to LeCun,[37] who placed learned world models at the center of a vision for autonomous machine intelligence. Today, the term spans a wide range of uses and meanings across domains.[9,11–13,38–41] In Box 1, we detail how world models have been developed in video games, robotics, and autonomous driving, which provide useful contrasts to their emerging role in biomedicine.

Components of a biomedical world model

Following the use in reinforcement learning, we define a world model by the central capability it provides rather than through a particular architecture or training objective. A biomedical world model is an AI system that generates distributions over how the state of a biological system may evolve under alternative actions and that can continue simulation from the states it predicts (Figure 1A).[10,37] Given an initial state, a world model can be manipulated by applying different actions to predict the resulting state of the system and use that predicted state as the basis for further simulation. This ability to generate action-conditioned, temporally extended, and branching predictions makes world models useful for planning toward specified goals.

We use the term *action* for the controllable input to the model. Genetic perturbations, environmental exposures, chemical treatments, clinical therapies, and experimental manipulations are domain-specific forms of actions. We use *intervention* for the corresponding manipulation of the biological system in the real world, particularly when discussing its causal effects. This distinction is important because a model can be conditioned on an action without necessarily having been trained on data that would support inferences about what would happen if that action were experimentally imposed.

We consider three components of biomedical world models: state and observations, actions, and biological dynamics.

State and observations

At time t, a cell, tissue, organism, or patient has a biological state x_t. This state is generally only partially observed, as not all modalities from atomistic to patient scales can be observed by a biological experiment (Figure 1A). Let $o_t \sim p(\cdot|x_t)$ denote a

stochastic observation process, where o_t are observations collected at time t, and $o_{\leq t}$ are observations up to that time. We use s_t to denote model state, which is information carried forward to predict how the system may evolve under future interventions. The model state may consist of current observations, the complete observation and intervention history, a sequence of tokens, modality-specific representations, explicit mechanistic variables, learned latent variables, or a combination of these. For world models that represent biology in a latent space, an encoder E transforms information available up to time t into a model state s_t. Specifically, E can combine observation history o_t, prior actions $a_{\leq t}$, and contextual information c into a latent representation that carries forward information needed for future prediction. A decoder D can then map the resulting state s_t back to a distribution over observable measurements, allowing predictions made in the latent space to be related to quantities that can be experimentally measured. The encoder-decoder framework provides one way to implement a world model, but neither is required under our definition.

Because the available observations may be compatible with multiple biological states, given by the posterior $p(x_t|o_{\leq t}, a_{\leq t}, c)$, a world model need not directly infer a single x_t. Instead, its state and predictions need to preserve any uncertainty about unresolved aspects of the biological system. This property is called predictive sufficiency: s_t should retain any information in the available history relevant to the distribution of future outcomes under possible actions, rather than enabling reconstruction of every unobserved feature of the system.

Actions

Biomedical actions differ in identity, for example, which drug or genetic perturbation is applied, as well as in dose, timing, duration, route, and combination. We use a_t to denote the action representation supplied to the dynamics model. Its representation may incorporate molecular sequence or structure, targets, dose, timing, duration, route, exposure schedules, or other properties of the intervention. The consequences of an action may depend strongly on biological state, but that context is represented by the current state s_t, rather than by the action itself. Whatever its form, the action must remain externally specifiable so that the same pre-action state can be queried under alternatives.

In some models, an action may be implemented as an explicit, user-controlled modification of designated state variables rather than as a separate action token. A germline perturbation, for example, could be represented by setting genotype before simulating development.

Biological dynamics

World models represent dynamics through an action-conditioned update, $s_t \sim T_\theta(\cdot|s_t, a_t, \Delta)$ where T_θ is a distribution over the next model state after elapsed time Δ. Depending on the architecture, this update may predict future measurements directly, append generated events to an autoregressive history, or evolve explicit or learned latent variables. Because s_t is constructed from partial observations, the predicted distribution may reflect both uncertainty about the current biological state x_t and stochasticity in the subsequent evolution of states. Repeated application of this update supports simulation over multiple steps and sequences of interventions (Figure 1B):

$$S_{t_{k+1}} \sim T_\theta(, |s_{t_k}, a_{t_k}, t_{k+1} - t_k), k = 0, \ldots, k - 1. \quad \text{(Equation 1)}$$

These capabilities define computational scope of a world model, but they do not establish that its predictions reflect the consequences of interventions applied to a biological system. That depends on the inferences supported by the data. Actions observed in observational data may be confounded by unobserved aspects of biological state, destructive measurements may leave transitions between individual units of a biological system unresolved, and predictions for states or actions not represented in the data require generalization beyond direct experimental evidence. These issues recur throughout biomedical world modeling. We return to them below when discussing the data needed to learn biological dynamics, the assumptions and architectures used to model them, and the experiments needed to establish predictive validity.

Related classes of models

We discuss several classes of models and how they relate to biomedical world models, focusing on whether a model represents the state of a system, accepts a specified action, predicts the resulting state, and can continue simulation from that predicted state.

Supervised and generative models

Prediction and generation are necessary capabilities of world models, but most predictive and generative models are not world models. A supervised model may map a rich biological state to a narrow target, such as predicting cell type from a transcriptomic profile. A generative model may perform the reverse mapping, for example, by sampling a transcriptome conditioned on a cell type label. Neither model represents a transition between biological states. For a predictive or generative model to function as a world model, both its input and output must represent wide system states, and the transition between them must depend on a specified action. For example, a diffusion model that takes a current transcriptomic state and a perturbation as input, then samples the resulting transcriptomic state, can serve as a world model if its predicted state can be used as the starting point for subsequent simulation.

Foundation models

Foundation models are pretrained on large and diverse datasets to learn representations or generative capabilities that can be transferred across tasks and contexts.[42,43] Models such as AlphaGenome,[44] Evo,[45,46] and Nucleotide Transformer[47] learn representations of DNA from variant to sequence to genomic scales; ESM-3,[48] ESM Cambrian,[49] and ProTrek[50] learn representations of protein sequence, structure, and function; Uni-Mol[51] and ATOMICA[52] learn representations of molecules and intermolecular interactions; Geneformer,[53] scGPT,[54] and UCE[55] learn representations of transcriptomic state; UNI,[56] CONCH,[57] and Prov-GigaPath[58] learn representations of pathology images and slides; VirTues[59] and TERRA[60] learn representations of tissues; and COMPASS[61] and PULSAR[62] learn representations of multicellular systems and patients. Regardless of scale, multimodality, or generative capacity, a foundation model only qualifies as a world model if it supports action-conditioned next-state prediction. However, these foundation models

may still serve as components of a world model, including encoders or decoders of states or interventions.

Mechanistic models

World models need not be learned from data. Biological dynamics can be modeled using equations, rules, constraints, or interactions derived from established physical, chemical, or biological knowledge. Such mechanistic models are world models when they can be initialized in a specific state, subjected to an intervention, and propagated forward to predict future states. This category includes ordinary and partial differential equation models, stochastic and agent-based simulations, whole-cell models, and many pharmacokinetic and pharmacodynamic models.[63,64] Although, in this perspective, we focus on learned world models, many world models may incorporate both learned and prespecified attributes. Such hybrid world models could encode established conservation laws, reaction networks, anatomical constraints, or pharmacological mechanisms while learning uncertain parameters, unresolved interactions, or other dynamics from data.

Agentic environments and virtual simulators

Large language models were originally developed as chatbots, but with advances in agentic AI, they can act on external resources through an agentic harness, the computational environment in which a model runs.[14] These harnesses provide tools that the model can call, agentic skills that combine analyses into repeatable workflows the agents can execute, and memory modules that store past analyses and can recall them later.[65] Agentic environments populated by generative agents that mimic human behavior[66] can simulate laboratories, clinical encounters, and research teams with the transitions between simulated states following a language model's prior over plausible continuations.[67–69] Emerging agentic laboratories separate a representation of laboratory state, which predicts the consequences of physical actions such as transferring a sample, from a representation of the system, which predicts how that system evolves under experimental intervention.[70] Agentic environments could serve as components of a world model by simulating the effects of experimental interventions and grounding these simulations in the state of the physical laboratory where experiments are performed.

Digital twins

Originally derived from the aerospace industry, digital twins is used as an umbrella term for virtual replicas of physical systems, such as a specific cell line, organoid, experimental system, patient, or laboratory process.[71] By contrast, world models can describe generic or population-level dynamics. Thus, a model of drug response across time may be a world model without being a digital twin. If that same model is fine-tuned using measurements from an individual patient, it may become a digital twin for that patient. Conversely, a digital twin that only mirrors the current condition of a physical system is not a world model, as it does not support intervention-conditioned next-state prediction.

DATA FOR BIOMEDICAL WORLD MODELS

Data requirements and current limitations

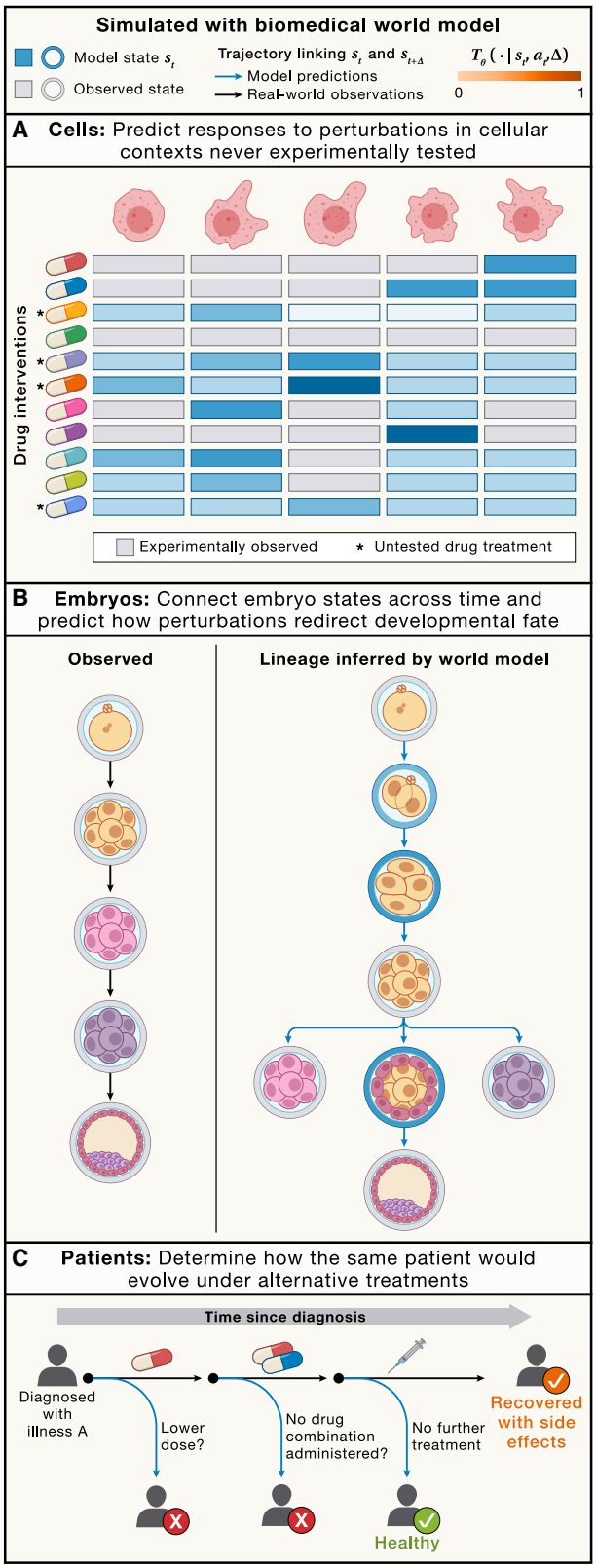
Training biomedical world models requires data that collectively provide information about the state of a system, the intervention applied, the resulting state, and the context in which the transition occurred (Figure 3A). This information about biological dynamics need not come from a single collection of trajectories. Observational data can capture the diversity of biological states and their natural evolution, while controlled experiments provide the causal evidence needed to learn how interventions alter those dynamics. Longitudinal and repeated measurements can reveal how biological changes unfold over time and, when trajectories of individual units are important, establish correspondence between observations at successive time points. No current data source provides all of this information across a sufficiently broad range of states and interventions.

Ongoing efforts to build world models of cells, embryos, and patients illustrate both the requirements and the limitations of available data sources. We focus on these as representative case studies rather than an exhaustive survey, which might examine world models at other biological scales such as proteins, tissues, or organs.[49] Single-cell perturbation screens observe controlled genetic and chemical interventions across a relatively small number of cellular contexts.[72–74] Developmental atlases measure biological change across time and space, but the most comprehensive molecular measurements are often destructive to the sample, precluding correspondence between cells across time.[75–77] Longitudinal health records follow the same patient across months or years; however, treatments are selected rather than randomized, measurements are irregular, and confounding variables are unobserved.[78–80] We explore these case studies as illustrations of the opportunities and challenges in data generation for biomedical world models.

Cellular world models

Virtual cells are computational models that aim to mimic the behavior of human cells.[81] Recent virtual cell models, such as GEARS,[82] CellFlow,[83] STATE,[84] TxPert,[85] STACK,[86] X-Cell,[87] and MORPH,[88] are often trained on large-scale single-cell RNA sequencing (scRNA-seq) or CRISPR-based Perturb-seq atlases,[72,73] accept one or a population of cells as input, and predict the transcriptional impact of a given perturbation. Because perturbations are experimentally assigned, these datasets can provide causal information about population-level responses within the cellular contexts they cover. However, the measurements are destructive, so they do not reveal how individual cells evolve under perturbation. Virtual cells, therefore, often carry the population of cells as the *state*, without requiring an identified cell-to-cell coupling. When this predicted output population can serve as a valid starting state for further iteration, these models are aspirationally cellular world models (Figure 2A).

However, virtual cells have faced substantial generalization challenges and are often outperformed by simple linear baselines,[89–93] especially for predictions about cellular contexts not well-represented in the training data. The poor generalization of virtual cells is reflective of the sparse coverage of cellular state and intervention space by existing perturbation datasets,[94] especially as perturbation effects are strongly dependent on cellular context.[95] Genetic knockouts have only been exhaustively categorized in a small set of cell types,[72] and although some chemical perturbations have been profiled more broadly,[73,74] exhaustive coverage of the vast chemical space has remained infeasible. Furthermore, beyond perturbagen



identity, the interventional effect also depends on dosage, timing, duration, and other elements of the experimental protocol, which are rarely recorded in harmonized, machine-readable formats and are inconsistent across datasets. Lastly, measuring the effect of genetic and chemical perturbations on transcriptomic state offers only a narrow view of cellular response and behavior: perturbations also affect protein abundance and localization, epigenetic state, signaling, morphology, differentiation, and intracellular and extracellular interactions, leaving much of the biologically relevant state space inaccessible to current cellular world models.

Embryonic world models

Virtual embryos aim to model embryogenesis as an evolving multicellular system whose state might include transcriptomics, morphology, spatial organization, and lineage information.[96] These aspirational world models accept the current embryonic state as input and predict the next state (Figure 2B), but as of yet usually only support actions implicitly, for example, simulating germline defects by manipulating the initial state. Mammalian embryonic atlases measure biological change across diverse developmental stages, tissues, contexts, and timescales, but unlike Perturb-seq screens in cell lines, they overwhelmingly follow natural development.[75,76,97–99] Interventional datasets are beginning to emerge, but, while promising, they remain narrow in perturbational breadth, particularly in mammals.[100–102] In the absence of large-scale perturbation screens in embryos, accurately reproducing normal embryogenesis does not establish that a model can predict the consequences of developmental interventions since these datasets may not satisfy the necessary causal assumptions.

The lack of available perturbation data is further complicated by destructive measurement. Bulk and scRNA-seq destroy the embryo, so developmental atlases observe different cells at successive stages instead of following the same cells through time. The fundamental question in developmental modeling is how cells differentiate, divide, and die across these states, yet population snapshots alone do not uniquely determine the dynamics connecting them.[77] A broad literature has therefore sought to reconstruct aspects of developmental dynamics by introducing additional structure. Pseudotime approaches such as Monocle[103,104] and Palantir[105] infer progression, branching, and fate structure from variation across cells; RNA velocity[106] and scVelo[107] use transcriptional kinetics to estimate local directionality; Dynamo[108] reconstructs transcriptomic vector fields; and optimal transport approaches such as Waddington-OT[109] and moscot[110] infer couplings between populations observed at different times. However, the inferred cell-cell correspondences depend on modeling assumptions, and transcriptomic state alone has been shown to be insufficient to identify true developmental ancestry.[111]

Figure 2. World models enable action-conditioned reasoning across cells, embryos, and patients

(A) Cellular world models can predict responses for perturbation-context combinations that have not been experimentally measured.

(B) Embryonic world models can connect developmental states across time and infer how perturbations redirect lineage and fate trajectories.

(C) Patient world models can simulate alternative treatment trajectories for the same patient and compare their predicted outcomes.

Spatial transcriptomic methods like seqFISH,[112] sci-Space,[113] Slide-seq,[114] Stereo-seq,[115] and whole-embryo MERFISH[116] enrich these snapshots by preserving the spatial organization of gene expression but still require terminal processing through tissue fixation, freezing, or sectioning, preventing repeated measurements of the same developing system. Lineage tracing methods provide additional information about ancestry even when molecular measurements are destructive. Heritable barcodes can identify clonal relationships among cells or, when lineage information accumulates across successive divisions, support reconstruction of more detailed lineage trees.[117–119] These genealogical measurements can then be combined with molecular state to constrain the developmental transitions connecting observations across time.[120] However, lineage relationships alone generally do not reveal the complete molecular trajectory followed between observations. Live cell imaging methods like *in toto* confocal and light-sheet microscopy enable direct observation of lineage, morphology, and spatial position,[121–123] but spectral constraints on fluorescent reporters limit measurements of molecular cell state to relatively few features.[124] No existing datasets simultaneously capture lineage-resolved trajectories, deep multimodal cell and tissue measurements, and systematic perturbations across development. Integrating these technologies will be necessary to train and evaluate embryonic world models.

Patient world models

Unlike destructive cellular assays, electronic health records provide linked sequences of medical events from the same individual over time and across populations. Generative medical event models, such as Foresight,[125] ETHOS,[126] Delphi-2M,[78] Curiosity,[127] and Apollo,[128] tokenize encounters, diagnoses, medications, lab measurements, procedures, and other medical events; then, they leverage autoregressive generation to predict or sample successive events conditioned on a patient's history. Because accumulated event history can serve as an evolving state, these models can simulate long-horizon patient trajectories and are thus aspirational world models (Figure 2C). They can also generate action-conditioned patient trajectories by modifying the patient history, for example, by adding a treatment, diagnosis, or procedure.[129,130]

Counterfactual validity, however, is difficult to establish from observational health records since they reflect both patient biology and interactions of the patient with the healthcare system. Observations are shaped by heterogeneity in clinical decision-making, access to care, institutional workflows, coding practices, billing requirements, and other sources of selection and measurement bias.[131–133] For example, a diagnosis code may indicate diagnostic suspicion or testing rather than confirmed disease, and its first appearance reflects when the disease was first documented or coded rather than the onset of pathology. Consequently, individual events can have low predictive value: for instance, 17.7% of patients with an ICD-10 code for pulmonary embolism[134] and 44% of patients with three or more ICD-9 diagnosis codes for rheumatoid arthritis do not actually have the disease.[135] Although health records offer long patient trajectories, they only provide a confounded and partially observable view of the state transitions that patient world models seek to learn. A model trained to reproduce these trajectories may learn the dynamics of healthcare delivery rather than how the patient would respond to an intervention.[78]

These examples suggest that no single data source captures everything needed to learn the complex dynamics of biological systems. Perturbation screens provide causal evidence but cover a small number of states and actions; developmental atlases capture natural dynamics but often lack temporal correspondence and contain few interventions; and health records provide longitudinal trajectories but include confounded actions and only partial observations of human disease. A practical path forward is to combine broad observational data with controlled longitudinal experiments, which involves generating multimodal datasets that capture a wider range of interventions and preserve biological information over time.

Generating interventional and trajectory data

Realizing the vision of biomedical world models will necessitate not only the use of existing data but also substantial investment to create new datasets for intervention-conditioned prediction (Figure 3B). The most direct supervision for biomedical world models comes from linked measurements of biological state, intervention, and resulting state across sufficiently diverse contexts, as well as longitudinal sequences of interventions applied to the same system (Figure 3D).

Non-destructive measurements

Advances in measurement technologies could enable less destructive and more repeatable assays, allowing the same system to be assayed before, between, and after sequences of interventions. Some methods directly sample intracellular material with minimal disruption; for example, fluidic force microscopy-based Live-seq,[136] nanotweezers,[137] or nanopipettes integrated with scanning ion conductance microscopy[138] can biopsy cytoplasmic RNA, enabling repeated molecular profiling while preserving cell viability. Other techniques nondestructively export molecular information for extracellular measurement; for instance, COURIER[139] and NTVE[140] induce export of RNA cargos or barcodes in virus-like particles or protein nanocages, allowing longitudinal measurements of clonal dynamics or transcriptomic state. Raman microscopy can measure vibrational signatures of the same living cells over time, with Raman2RNA[141] demonstrating that these label-free Raman spectra can be mapped to single-cell expression profiles. Still, other approaches do not read out the cellular state while the cells remain alive; however, they intracellularly record information about cell state for later recovery. For example, TimeVault[142] and ENGRAM[143] record transcriptional state or pathway-specific signals in a cytoplasmic ribonucleoprotein complex or prime editing-based DNA memory device, respectively. Although these methods cannot yet achieve the throughput or molecular depth of destructive profiling methods, they illustrate a path toward generating longitudinal molecular trajectories for biological world models.

Organoids and related model systems

Although perturbation and longitudinal molecular measurement are often challenging in humans, improved models could provide accessible settings to generate such data while retaining key features of human biology. For example, while the living human brain is largely inaccessible to repeated molecular

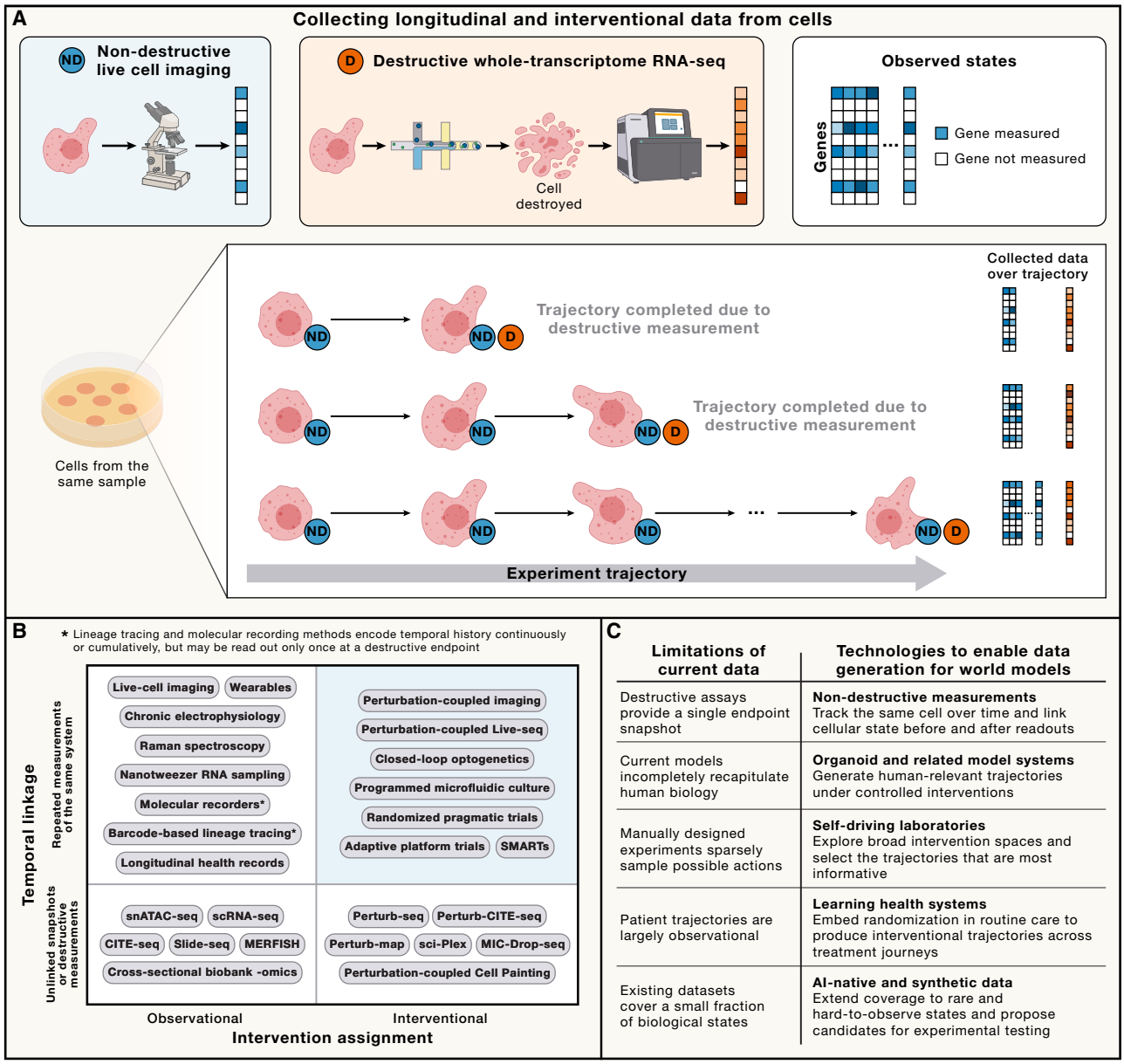


Figure 3. Data for biomedical world models

(A) Repeated non-destructive measurements can track biological systems over time, whereas destructive assays provide richer endpoint measurements but terminate trajectories.

(B) Representative biomedical data sources and measurement technologies vary in intervention assignment and temporal linkage.

(C) Emerging technologies and experimental systems could address limitations in longitudinal measurement, human relevance, intervention coverage, causal patient trajectories, and biological state coverage.

measurement, long-term live imaging and single-cell transcriptomics of brain organoids have enabled tracking of morphogenesis, membrane and cytoskeletal dynamics, and tissue-state transitions over days to weeks[144]; other platforms have enabled chronic electrophysiological recording of individual organoids for up to 4 months.[145] Organoids have also enabled high-throughput characterization of responses to genetic, morphogenetic, and combination perturbations; for example, in neurodevelopment,[146–148] breast cancer,[149] and gastric cancer[150] models. Assembloids and multidonor chimeroid models extend these experiments to broader contexts, allowing tissue or donor-specific responses to be studied at scale.[147,151] To create multi-organ world models, multiple organ-on-a-chip models can be fluidically coupled to simulate properties arising from interactions between organ systems, such as pharmacokinetics and pharmacodynamics.[152] These human-body-on-chip systems can be robotically maintained and interrogated for longitudinal data generation.[153] At still

greater physiological fidelity, *ex vivo* organ perfusion systems preserve the function of human organs while permitting biosampling, imaging, and intervention to create whole-organ digital twins,[154] and brain-dead decedent models may even extend this paradigm to the organism level.[155,156] Together, these models demonstrate how interventional trajectory data could be produced across progressively richer levels of human biological organization, fidelity, and context.

High-throughput self-driving laboratories

Advances are also needed in how data generation experiments are selected and executed. Programmable cloud laboratories[157,158] and biofoundries[159] could execute experimental protocols translated into machine-executable workflows that can be robotically scaled, enabling broader coverage of the intervention space. Furthermore, agent-driven autonomous laboratories could directly integrate both world models and the ground-truth systems they represent, or models thereof, to enable rapid closed-loop design-build-test-learn cycles.[160–162] An agentic controller can query the world model to generate predictions, interface with robotic infrastructure to experimentally test these predictions in the ground-truth system, then update the world model with the results. Rather than exhaustively enumerating intervention space, the controller can implement active learning strategies to prioritize interventions that reduce uncertainty, discriminate between competing models, or improve coverage of poorly sampled biological states. Active learning and Bayesian optimization strategies have already been shown to facilitate navigation of vast combinatorial spaces, including for protein engineering,[21,163–165] lipid nanoparticle design,[166] and drug combinations,[167] and accelerate development of systems biology models, for example, of the yeast diauxic shift[168] or CXCL9 regulation in pancreatic cancer cells.[169] These modeling successes suggest that similar closed-loop strategies could iteratively improve biomedical world models. Lastly, data-generation platforms should retain not only successful experiments but also failures, null results, toxic conditions, and protocol deviations to avoid datasets shaped by publication or human selection bias.

Learning health systems for patient trajectories

For patient world models, observational health records should be supplemented with both richer molecular measurements and prospective and interventional data generated within learning health systems.[170,171] Large-scale biobanks provide infrastructure for linking health records to rich molecular measurements, including genomic variation, plasma proteomics, metabolomics, and other molecular phenotypes.[172,173] Randomized trials can complement these observational data with causal evidence about how patient states change under intervention but are constrained by feasible populations, interventions, and follow-up windows. Electronic health record-embedded pragmatic trials[174] and registry-based randomized trials[175] demonstrate that routine clinical infrastructure can be used to randomize treatment and scalably ascertain longitudinal outcomes. Adaptive platform trials can evaluate multiple treatments, combinations, and patient subgroups under a shared operating protocol,[176] while sequential multiple-assignment randomized trials assign randomized interventions at successive decision points,[177] directly generating sequences of patient state, intervention, and subsequent state. Where randomization is not feasible, target trial emulation can support generation of real-world evidence by applying causal inference to estimate treatment effects.[178,179] Although some examples of high-throughput trial emulation exist,[180] they remain challenging to scale and vulnerable to unmeasured confounding, data errors, informative observation, and violations of positivity.[181,182] Moreover, such studies typically estimate population-level treatment effects rather than generating individual-level trajectories of patient state, intervention, and response. Generative models combined with new trial designs could facilitate scalable trial emulation, providing causally informative longitudinal outcomes for training patient world models.[183–186]

Synthetic data

Although they inherit the assumptions and biases of the data generating process, synthetic data generated by mechanistic simulators, statistical models, or learned generative models could complement experimentally observed trajectories.[187] Existing approaches range from mechanistic simulations of multicellular systems[188] to statistical generation of single-cell and spatial trajectories[189] or longitudinal clinical time series.[190] These methods can expose models to states, interventions, temporal resolutions, or rare events that are difficult, costly, or unsafe to observe experimentally.

AI-native data and interoperability

Biomedical datasets are often designed around human-readable outputs rather than machine learning. High-dimensional images, spectra, waveforms, and other raw signals may contain predictive information that is discarded when measurements are reduced to derived features or summary statistics. Preserving the original minimally processed observations could improve predictive performance of world models trained on these datasets.[191] Datasets should also be interoperable across laboratories, institutions, modalities, and biological scales.[192–195] This will require standardized representation and recording of biological and environmental context, interventions, outcomes, instrument and batch effects, and lineage or subject identifiers, among other contextual variables. Raw and processed measurements should be linked to machine-readable metadata, executable experimental protocols, and analysis code, enabling datasets to be reproduced, integrated, and reused for world model training. Where data cannot be centralized due to privacy or governance constraints, standardization could enable federated training of biomedical world models.[196]

TRAINING BIOMEDICAL WORLD MODELS

Training a biomedical world model requires solving several key problems: constructing a predictive model state from partial observations and history, representing actions so their effects can generalize across states and contexts, and learning how biological states evolve under those actions (Box 2). Different data sources provide different information about these dynamics. Broad observational data capture biological states and their natural evolution, controlled experiments reveal how actions alter those dynamics, and longitudinal or repeatable measurements link states through time. We first consider how these data can be used to construct the model state and action

Box 2. Design principles for biomedical world models

These principles describe functional requirements for biomedical world models that support iterative simulation and planning.

Closed predictive state: The model state should preserve the information needed to predict future behavior under possible actions. For example, transcriptionally similar cells may require different model states if their histories imply different responses to the same perturbation. Given a model state and an action, the transition model should be able to sample a distribution over updated model states that can be used for further simulation.

Structured action representation: A world model needs a representation of each intervention it may simulate. Rather than assigning every intervention a categorical identifier, the model should encode properties that describe its biological effects. For compounds, these could include chemical structure, molecular targets, dose, and mechanism of action. For genetic perturbations, they could include target sequence or structure, perturbation type, and direction of effect. Structuring the action space around biological properties allows the model to share information across related interventions and represent new perturbations or combinations that were not encountered during training.

Biologically expressive dynamics: Transition models should capture several complexities that are especially important in biomedical systems. Biomedical dynamics are often stochastic, so models should represent distributions over possible future model states rather than a single outcome. Biological states may also contain many interacting units, such as spatially adjacent cells within a tissue, requiring architectures where units can influence each other's evolution. Finally, architectures may need to accommodate compositional and organizational changes to the underlying system, for example, as cells divide, die, and migrate.

Reliable multi-step simulation: Accurate one-step prediction does not ensure reliable simulation over multiple steps. Repeated application of the transition model should remain stable: trajectories generated from the model's own predicted states should remain biologically plausible and preserve the diversity of possible futures rather than drift or collapse. Predictions over the same interval should also agree whether generated directly or by composing shorter transitions.

Limits of model design: Even if a biomedical world model satisfies the preceding design principles, the dynamics it represents may not actually be identified from the available training data. In observational settings, multiple causal dynamics can be compatible with the same observed data; with destructive measurements, multiple unit-level transition processes can likewise be compatible with the same sequence of population snapshots. Beyond model design alone, addressing these gaps in identification requires new data that provide the missing information, together with modeling approaches that can exploit it.

representations, then how observational, interventional, and longitudinal information can be combined to learn action-conditioned dynamics across time and biological scales.

Learning representations of states and actions

Biological state

A first challenge is to learn representations of biological state. Foundation models pre-trained on large-scale genomic, single-cell, imaging, or other modality-specific datasets can provide valuable starting points for these representations.[44–62] Rather than learning state encoders from the limited datasets that contain interventional trajectories, world models can initialize encoders from foundation models and subsequently adapt them to prediction of biological dynamics. For example, scGPT was pre-trained on 33 million cells, then fine-tuned for one-step perturbation prediction.[54] Chreode pre-trained an scVI encoder on 2.48 million cells from a mouse embryonic atlas; the resulting frozen scVI embeddings served as state representations for a transition model.[197] While pre-training does not guarantee that representations preserve the information needed for intervention-conditioned dynamics, it provides a data-efficient way to learn state structure from much broader observational data before adapting the representation to world-model prediction.

Even with strong pre-trained encoders, biology remains partially observed, and different measurement modalities reveal complementary aspects of biological state. For world models, multimodality is especially important because the richest molecular assays are often destructive, whereas other modalities can repeatedly measure the same biological system over time. Integrating these modalities can therefore both enrich state representations and link deep biological measurements to longitudinal trajectories. When modalities are jointly observed, techniques such as contrastive alignment,[57,198,199] masked reconstruction,[54,200] or cross-modal generation[201,202] can allow observed modalities to inform those that are missing. MultiVI, for example, jointly models RNA and chromatin accessibility,[203] while MIDAS extends this idea to mosaic datasets containing different combinations of RNA, chromatin accessibility, and protein measurements.[204] When paired data are unavailable, correspondences between modalities can be inferred using known biological relationships, for example, that gene expression measurements in scRNA-seq data and spatial transcriptomics should correspond.[205–207] Bridge integration uses a smaller multimodal dataset measured in the same cells to connect larger unimodal datasets.[208] Modality-specific representations can also be aligned to the embedding space of a language model, using text as the anchor to link heterogeneous multimodal observations.[209–213] Diffusion and flow-based models can treat missing modality inference as an inpainting problem, flexibly integrating partially observed datasets; however, they cannot infer cross-modal correspondence without linking information.[214–217]

These approaches can provide richer views of biological state and, where temporal bridge measurements are available, key information linking state at different resolutions across time. Model state construction then accumulates this information across the observation and intervention history o_t, $a_{\leq t}$, together with context t, into the model state s_t used for future prediction. The model state s_t must preserve the information in the available observation and intervention history that is relevant to future behavior. This model state can be learned separately or end-to-end from the multimodal integrative models above and the biological dynamical models discussed below. This motivates predictive representation approaches, such as joint-embedding predictive architectures (JEPAs) like I-JEPA[218] and V-JEPA,[12,219] that model future representations rather than reconstructing every detail of the observations.[37,220] JEPA and related approaches may be attractive in biomedicine to

discard assay noise or fine-scale imaging details that are irrelevant to subsequent behavior.

Actions

An action representation provides a manipulable input to the dynamics model. Its central requirement is to encode the properties of an intervention that distinguish biologically different consequences. Biomedical interventions are often highly structured: drugs can have different effects depending on dose or duration, genetic perturbations can be transient or persistent, and multiple interventions can be applied simultaneously or sequentially. Their effects may therefore depend not only on which interventions are applied but also on their intensity, timing, interactions, and order.[221,222] The action representation should capture these properties, while the dynamics model determines their state-dependent consequences after being applied to a specific state. Interventions can also comprise multiple actions applied simultaneously, such as combinations of drugs or genetic perturbations. Set-based architectures can represent these unordered combinations while learning interactions among their components.[223,224]

Because only a small fraction of the action space can be measured experimentally, generalization to unmeasured actions or state-action combinations requires learning or imposing structure on the action space. This structure can come from several complementary sources: biological and medical foundation models can provide pretrained representations of genes, proteins, molecules, treatments, and their interactions[44–52]; functional databases like DepMap and the Connectivity Map can characterize perturbation effects across contexts[74,225]; and biomedical knowledge graphs can encode relationships among targets and pathways, as in the co-expression and Gene Ontology structure used by GEARS.[82,226] While learned structure can support extrapolation beyond directly measured interventions, it does not by itself identify their causal effects.

Modeling biological dynamics

Causal dynamics

The dynamics model must learn how biological systems evolve under interventions. A fundamental challenge is that interventional effects are not always identifiable from observational data. In clinical data, for example, treatments are often selected based on disease severity, which is not precisely recorded in the health record. Similarly, a gene whose expression change is associated with a cell state transition may not have caused that change since both might have been influenced by an unobserved regulatory factor. Causal interpretation requires assumptions such as no unmeasured confounding and positivity, which we know are unlikely to hold in many biomedical contexts. This does not necessarily mean observational data are not useful. Observational data can provide a kind of inductive bias over plausible intervention-conditioned dynamics through broad coverage of natural developmental trajectories. Controlled experiments can then provide causal supervision for how interventions behave differently from the natural evolution of states, reducing the amount that must be learned from sparse interventional data alone.[227] This strategy is useful only insofar as observational and interventional dynamics share transferable structure, which is difficult to verify, and without direct experimental support, predicted intervention effects remain generalizations that are possibly non-identified.

Unit-level and population-level dynamics

Learning dynamics also requires establishing how observations are connected across time. When the same system can be measured repeatedly, as in longitudinal patient records or live imaging, observations provide coupled evidence about the evolution of that system. Destructive assays, by contrast, measure the populations observed before and after a perturbation or at successive developmental stages, but not how cells observed at one time are related to those observed later.[77] In development, this relationship is not simply a one-to-one match: cells can divide or die before the next observation, further complicating the problem of inferring unit-level dynamics. If cell-level trajectories are the target, these correspondences must therefore be measured or inferred. They can be estimated using kinetic, geometric, or transport assumptions,[105–110] constrained by lineage information,[109,120,228] or observed more directly through repeatable measurements such as live imaging. Such measurements can also serve as temporal bridges linking trajectories of the same system to richer destructive assays through multimodal alignment.[136–140,142,143] When the unit-level dynamics are not identified, populations of cells can themselves serve as the model state, allowing the model to learn the distribution of states that follows an intervention without assigning each predicted cell to a particular input cell.[83,84,87,88,229]

Transition models

The transition architecture determines what kinds of dynamics the model can represent. A basic choice is whether state is maintained implicitly in a sequence or propagated explicitly in a learned state space. Sequence models, such as generative patient models, factorize trajectories into successive conditional predictions over observations or tokens.[78,125–128,230] Latent world models instead propagate learned model states without reconstructing the full observation at every step.[9,220,231,232] These approaches can also be combined, for example, by applying autoregressive prediction over learned latent states.

Whatever the representation, the stochasticity of biomedical systems means that model states and actions do not uniquely determine the next state. This kind of predictive uncertainty arises because partial observations provide incomplete information about the underlying biological system and because subsequent biological dynamics may themselves be stochastic. Transition models should therefore represent conditional distributions over future states rather than only point predictions. Stochastic latent variable[220] and autoregressive models,[230] diffusion,[233,234] and flow matching[83] approaches provide different mechanisms for representing these distributions. The goal is to preserve distinct possible outcomes rather than average them into a single predicted state.

Biomedical systems are often composed of many biological units, which has implications for how transition models must be designed. A cell's response to an action can depend on neighboring cells, population composition, spatial context, and extracellular signals. These kinds of interactions between units require architectures like set-based attention or graph-based approaches to provide mechanisms for conditioning each unit's update on the broader system rather than transforming units

independently.[84,224,229,235] Moreover, transition models may also need to represent changes in the composition and organization of the system. During embryogenesis, for example, cells divide, die, and migrate, changing both which cells are present and their lineage and spatial organization. A transition model that only updates the attributes of a fixed set of units cannot represent these events; it must also be able to represent the creation or removal of units and changes in their organization.[234]

Learning dynamics across horizons and scales

Biological dynamics span heterogeneous timescales. A world model may need to connect rapid molecular signaling to slower changes in cell identity, tissue remodeling, disease progression, or therapeutic response. For example, EGFR tyrosine kinase inhibition in EGFR-mutant lung adenocarcinoma may suppress oncogenic signaling and produce short-term tumor regression, yet surviving cells may acquire drug resistance, leading to long-term disease recurrence and progression.[236] Predicted trajectories of EGFR inhibition should, therefore, capture both effects. The simplest approach to generate long-horizon rollouts is by recursively applying a one-step transition model, using each predicted model state as the starting point for the next simulated intervention.[9,231,232,234,237] This, however, creates a compounding error problem: even small errors at early time steps can accumulate and degrade the value of longer rollouts.[238–240] Rather than requiring every future state to be reached through repeated transitions at the same temporal resolution, models can instead represent time and prediction horizon explicitly in the dynamics. Continuous-time models provide established approaches for predicting across arbitrary intervals.[241,242] Long-horizon predictions can also be supervised directly. PlaNet, for example, trains latent predictions several steps ahead, while sequence and diffusion models can model distributions over complete trajectory segments.[220,230,233] As multi-horizon reasoning is essential for long-range planning and efficient inference, learning dynamics coherently across prediction horizons remains a key open problem.

Unlike some other domains where world models have been deployed, biology has interpretable multiscale structure. Higher-level biological states are often collections of lower-level states: a patient sample might contain a heterogeneous distribution of cells, and tissues contain spatially organized populations and microenvironments. In single-cell modeling, hierarchical architectures increasingly transform sets of cells into aggregate embeddings that capture properties of the underlying cellular distributions.[83,84,229,243,244] At tissue scale, aggregation must also preserve spatial relationships. Spatial graphs and hierarchical tissue models can construct representations of cells, neighborhoods, niches, and whole tissues, while foundation models such as VirTues and TERRA increasingly learn representations across several such resolutions.[59,60,235,245] Emerging model systems described above provide a supervised route to learning these relationships: organoids or assembloids can couple molecular and cellular measurements to tissue morphology and functional phenotypes within the same system,[144,246–248] while linked organ-on-a-chip platforms can model inter-organ interactions.[152,153] Such systems could provide paired trajectories across scales rather than requiring cross-scale relationships to be inferred from separately collected datasets. Overall, these advances suggest that multiscale state representation is becoming increasingly tractable; however, learning intervention-conditioned dynamics that operate coherently across these levels is still an unsolved challenge.

Temporal and biological hierarchies are ultimately coupled. Rapid molecular and cellular responses may require fine-grained simulation, whereas long-term consequences can sometimes be simulated through coarser, population, patient, organ, or tissue states. However, coarse variables should not simply replace fine ones: population or tissue context can feed back to alter cellular dynamics, and the same macroscopic state may be compatible with many microscopic configurations. Multiscale world models will require stochastic mappings between levels, bottom-up aggregation and top-down disaggregation, and consistency objectives requiring predictions made at a coarse scale to agree with those obtained by aggregating simulations of finer-scale dynamics. Models such as PULSAR begin to connect representations and perturbational predictions across cellular and multicellular resolutions,[62] but how to integrate multiple biological scales with multiple predictive horizons in a single coherent dynamical model is a frontier question in biomedical world modeling.

EVALUATING BIOMEDICAL WORLD MODELS

Evaluation of biomedical world models will require testing whether their action-conditioned simulations are valid as counterfactual predictions. This is an interventional question: given an initial state and an intervention that was not applied to that state, the model should generate the distribution of states that would result under the intervention. Establishing this requires testing generalization on held-out data, biological consistency of predicted responses, stable simulation over long horizons, and validation through comparing model predictions and actual experimental results (Figure 4).

Generalization benchmarks

The first bar for any world model is accurate prediction on held-out data (Figure 4A). These evaluations can be conducted without collecting new experimental observations. Benchmarks should evaluate generalization across biological states, such as new cell types, genotypes, donors, time points, or hospitals; across actions, such as unseen perturbations, doses, or treatments; across state-action combinations, in which the state and action have each been observed but not together; and across time horizons or intervention sequences.[93,109] Accurate assessment of generalization depends on appropriate construction of data splits into training and held-out test sets.[249–251] Randomly partitioning cells, measurements, or visits can place closely related samples from the same donor, lineage, patient, or trajectory on both sides of the split, leaking information from the test set into training and inflating model performance. Even when training and test samples are fully disjoint, the test samples may resemble the training data more closely than the states and actions the model will encounter in the real world, making the benchmark less

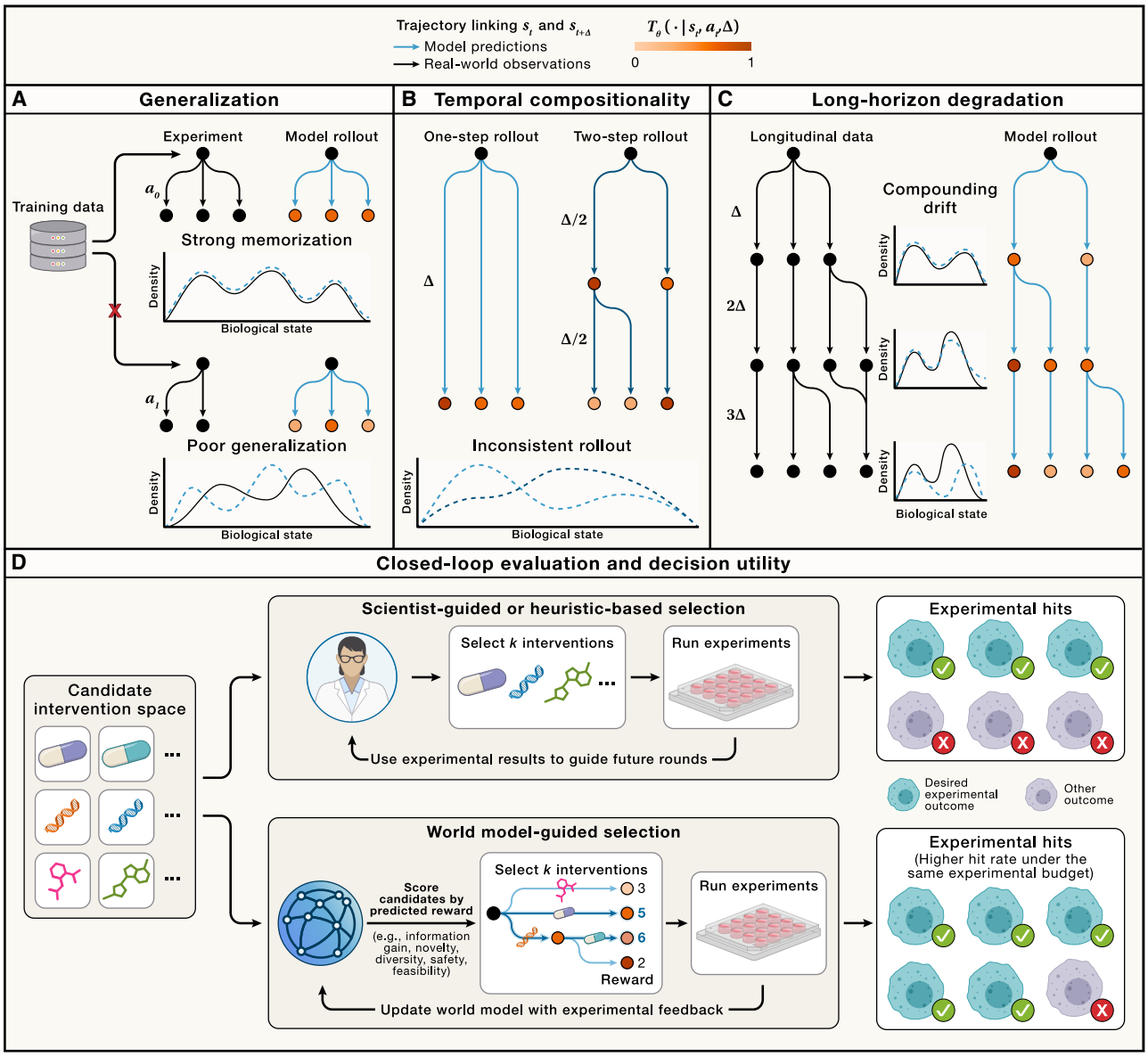


Figure 4. Evaluation of biomedical world models

(A) Generalization. Performance on observed states and actions can reflect memorization; evaluation should test held-out states, actions, and state-action combinations.

(B) Temporal compositionality. Predictions over an interval "Δ" should agree with predictions obtained by composing transitions over shorter intervals.

(C) Long-horizon degradation. Errors can compound when predicted states are reused for subsequent simulation, causing increasing distributional drift from longitudinal observations.

(D) Closed-loop evaluation and decision utility. World-model-guided experiment selection can be compared with scientist-guided or heuristic selection under the same objective and experimental budget.

informative of real-world performance. More stringent splits partition samples by biologically meaningful similarity; genetic perturbations can be split by gene family, pathway, protein complex, or interaction neighborhood; compounds by chemical scaffold, mechanism of action, or target family; developmental trajectories by contiguous time intervals or lineage branches; and patient data by patient, institution, geography, or time.

Comprehensive performance metrics

Comparing model predictions to unseen states also requires careful design of metrics. In perturbation prediction, the average expression difference between perturbed and unperturbed cells, a standard expression-based metric, is often explained by systematic biases shared across perturbations rather than by perturbation-specific effects.[92] A model that captures only the average perturbation effect can therefore appear to perform

well. Summary statistics such as gene-wise means, fold changes, variances, and covariances should be accompanied by distributional distances such as the maximum mean discrepancy or Wasserstein distance. Rank-based metrics assess specificity by testing whether a predicted response identifies its own perturbation among other candidate perturbations.[252] Predictive uncertainty should be evaluated alongside accuracy, with predicted probabilities agreeing with empirical frequencies and prediction intervals achieving nominal coverage. The metrics described here were developed for transcriptional readouts and apply to cellular world models; embryonic and patient world models predict spatial organization, lineage relationships, morphology, and the timing of clinical events and will require separate classes of performance metrics.

Baseline comparisons and ablations

Because model performance can arise from data leakage or systematic bias, it is essential to compare world models with strong, minimally structured baseline approaches. Such baselines are ablations of the world model, constructed by excluding individual components of the world model and measuring the performance of simplified models: stability baselines predict that the current state remains unchanged, natural dynamics baselines copy the current state forward and ignore the intervention entirely, pooled response baselines ignore the initial state and predict the average response to the action or the average response across actions, additive or linear baselines test whether modeling non-linear interactions is necessary, and nearest-neighbor or interpolation baselines predict intermediate states to test whether learned dynamics improve performance beyond nearby observed states. Perturbation models are often compared against control, average, additive, and linear baselines and frequently fail to exceed them.[89–93] Developmental trajectory models have been compared with linear interpolation between observed cell populations[253] and patient trajectory models with age- and sex-specific incidence rates, linear or logistic regression, gradient-boosted trees, and longitudinal statistical models.[78,125–128] Biomedical world models will need to be evaluated against similar baselines as well as against baselines that ablate the intervention and the initial state, which models discussed here do not represent.

Biomedical challenge tests

Biomedicine also contains a large body of established knowledge that a world model should capture and that can be used to evaluate its predictions (Figure 4A). Perturbation studies, lineage experiments, imaging and histology, mechanistic studies, and clinical trials each establish facts about how a system responds to an action; for example, that a knockout eliminates a cell type, a rescue restores a phenotype, a lineage transition cannot occur, or that a treatment acts in one patient subpopulation. World models can be queried in the same manner, and their outputs can be used to measure whether the model captures mechanistic and phenotypic consequences of interventions. For example, cellular world models can be tested on whether predicted responses recover non-additive genetic interactions,[82] reproduce morphological phenotypes,[254] or replicate cell state proportions and perturbation-induced gene programs,[255] while embryonic world models can be tested on whether simulated progenitors reproduce clone-specific fate probabilities.[253] We also intend this category to cover standard target trial emulation studies for population-level evaluation of counterfactuals from patient world models.[165–173] None of these tests can establish correct counterfactual trajectories, but they are important validations.

Representation quality

Beyond future predictive validity, models can also be evaluated by the information encoded in their internal state. Simple probes can test whether variables such as cell type, developmental stage, lineage, spatial context, disease severity, or treatment history can be recovered from their representations.[256,257] However, an embedding useful for retrieval, generation, or zero-shot transfer may discard distinctions that matter under intervention. For world models, a stronger criterion is predictive sufficiency. If $o_{\leq t}$ represents the observed history and s_t the internal state, then s_t should already preserve the information in $o_{\leq t}$ needed to predict future observations. For example, two naive $CD8^+$ T cells may have similar transcriptomic profiles but different T cell receptor specificities and therefore respond differently to the same peptide-major histocompatibility complex (MHC) exposure. The model state should capture these differences despite the superficial transcriptomic similarities. Predictive sufficiency can be tested by freezing the model and comparing matched next-state predictors that receive either s_t alone or s_t and $o_{\leq t}$. If adding the observed history improves held-out future prediction, the state representation has discarded information useful for future simulation. Interpretability can complement these tests by asking whether latent features correspond to known biological programs or mechanisms; however, human-interpretable representations are not necessary for biomedical world models. In fact, world models may exploit predictive structures that are not yet well understood.

Long-horizon evaluation

The evaluations discussed thus far primarily test whether one-step predictions hold. World models have an additional requirement: predicted states must remain valid inputs for further simulation. Free-running rollouts should therefore also be evaluated. As generated states are fed back into the model, small one-step errors can compound, producing increasingly implausible trajectories.[238–240] Models should be evaluated over progressively longer horizons and intervention sequences, comparing not only endpoint error but also distributional drift, loss of diversity, and violations of biological constraints. Many of these trajectory-level properties can still be assessed even when extended ground-truth demonstrations are unavailable, since they are effectively measures of degradation.

Temporal compositionality

Repeated simulation should also be temporally compositional. Consider the following Chapman-Kolmogorov-style test used to validate Markov state models.[258] For $i<j<k$, the direct predicted transition from s_i to s_k should agree with the distribution obtained by first simulating from s_i to an intermediate state s_j, then propagating the sampled intermediate states to s_k. A cellular world model could compare the distribution over

cellular states from directly simulating 48 h ahead with the distribution obtained by composing two 24 h transitions. Disagreement may indicate that learned transition horizons are inconsistent or that the internal state is not sufficiently Markovian and omits history required for continued prediction.

Cross-modal prediction

Similar consistency tests apply across measurement modalities and biological scales. If transcriptomics, imaging, electrophysiology, and spatial profiling provide partial observations of the same biological state, then they should induce a consistent distribution over future states, even if predictions from less informative modalities are more uncertain. Developmental models trained jointly on scRNA-seq and spatial transcriptomics could be initialized from either modality and evaluated on whether cross-modal predictions feature consistent cell type compositions or spatial organization. Predictions should also be consistent across scales; for example, aggregating simulated single-cell states should agree with tissue or bulk-level predictions.

Prospective validation

Retrospective benchmarks cannot establish predictive validity for interventions that have never been observed, so computational evaluation alone is not sufficient to determine whether a world model is useful. Establishing that requires prospective experiments in which the state, the intervention, the predicted outcome distribution, and the evaluation metrics are specified before the experiment is performed, and the prediction is then compared with the resulting observations (Figure 4A). Prospective studies follow the principles described above and deliberately target interventions, biological contexts, or intervention sequences that lie beyond the world model's training data. For patient world models, a strong prospective test is concurrent replication of ongoing clinical trials.[259,260] The model can be frozen before trial outcomes are observed and, using only information available at enrollment, generate patient-specific outcome distributions under each randomized treatment and control. Difficulty can sequentially increase, from a new molecule acting on a well-studied target to a new mechanism within a known pathway to a target with no prior human pharmacology.

Decision utility

Ultimately, biomedical world models should be judged by whether their predictions improve decisions about which interventions or experiments to pursue (Figure 4B). Stronger predictive performance does not imply better decision utility. Errors near a decision threshold carry disproportionate consequences, small absolute errors can reverse the ranking of candidate interventions, and large errors matter little when they leave the preferred action unchanged.[261] World model-based planning can be compared with one-step predictive models, greedy or Bayesian optimization, model-free reinforcement learning, expert heuristics, and other selection strategies under matched objective and experimental budgets. Evaluations can measure hit rate, enrichment, information gain, regret, and safety or feasibility violations. Closed-loop experimentation extends this comparison over multi-round evaluation. Such tests will evaluate world models holistically by decisions they contributed to rather than the accuracy of any individual predicted action or intervention.

MODES OF INQUIRY ENABLED BY WORLD MODELS

Across biomedical research, what potential advances could world models enable that would have the greatest impact? If the challenges previously enumerated can be addressed, biomedical world models whose action-conditioned simulations recapitulate interventional structure could (1) simulate the consequences of an intervention, (2) search for the intervention that best achieves a specified target, and (3) plan sequences of interventions toward a desired state.

Simulation

The most direct use of a causally validated world model is counterfactual simulation: specify an intervention and study how the system changes (Figure 1C). The value goes beyond the distribution of outcomes: biomedical world models can enable understanding how outcomes arise. For example, in development, a virtual embryo could generate rollouts under genetic perturbations to map how regulators alter cell-fate trajectories and attempt to identify gene regulatory programs governing differentiation. In regeneration, a tissue model could follow injury to understand why repair proceeds toward regeneration, fibrosis, or chronic injury.[262,263] Spatial world models could similarly ask how aging or rejuvenating interventions affect neighboring cells and tissue organization.[264] The central challenge is counterfactual fidelity: these simulations are useful to the extent they exhibit interventional structure beyond trajectories already observed.

Intervention design

Given a target state, world models could be *inverted* to search for an intervention predicted to produce that state (Figure 1D). Cellular world models could search for genetic perturbations or transcription factor combinations that induce a desired cell fate, or patient world models for a drug, dose, or combination that maximizes benefit while controlling toxicity. These action spaces quickly become large: existing drug-combination studies already confront millions of possible experiments, while organoid screens expose large gene-by-treatment spaces.[150,167] But world models, combined with algorithmic search, could prioritize promising regions of these spaces without requiring exhaustive evaluations. The primary concern in this kind of design setting is that optimization can exploit systematic errors in the world model, favoring actions that appear to achieve the objective because the model misrepresents their consequences, particularly under distribution shift. Planning must guard explicitly against violations of biological and safety constraints as well as against degenerate rollout behavior.

Sequential planning

Many biomedical objectives involve multi-step, path-based interventions, so optimizing a single step within a complex intervention is insufficient (Figure 1E). Cellular reprogramming, for example, can depend on the order and timing of transcription factors: sequential delivery of reprogramming factors

can outperform simultaneous exposure.[265] Cancer treatment provides another example of path dependence because each treatment changes the tumor state and can therefore alter how the tumor responds to subsequent therapies. A therapy changes tumor burden and also the composition of the surviving cell population, which can alter the response to subsequent treatments. Continuous treatment can select for resistant populations, motivating adaptive regimens in which treatment changes as the tumor state evolves.[266] More generally, adaptive clinical interventions formalize treatment as a sequence of decisions in which each choice depends on the patient's treatment history and current state.[267]

Here, optimization is over the space of trajectories rather than single actions. Even restricting to K candidate interventions over H decision points, exhaustive enumeration grows as K^H, with stochastic outcomes creating a still larger branching space. Planning, however, need not explore every branch. Model-based planners can sample promising trajectories, use learned value functions or heuristics to prune unlikely or low-value branches, optimize actions directly in a learned latent space, and plan hierarchically over temporally abstract states; tree-search and learned-policy approaches already make large planning spaces tractable in other world model settings.[9,268] Agentic controllers could combine these methods with biological constraints, uncertainty estimates, and external optimizers to decide which branches merit deeper simulation. Over longer horizons, hierarchical world modeling may become especially important. Models operating at coarser temporal or biological scales can first identify high-level plans and intermediate goals, while finer-scale, shorter-horizon models resolve the transitions needed to achieve them.[269,270] Over repeated simulation, model errors can accumulate, driving trajectories toward biologically impossible states, collapsing diversity, or pushing the model into poorly supported regions of state space. To ameliorate this issue, it may help to proceed in a receding-horizon fashion: simulate several steps ahead, execute one intervention, observe the resulting state, and replan. Each new measurement collapses part of the branching tree, providing a practical route to adaptive cellular engineering, treatment, and experimental design.

OUTLOOK

Biology has accumulated an extraordinary record of observed biological states and processes, while the far larger space of potential interventions remains largely unexplored. World models could make part of that space accessible through simulation. Reaching this point will depend on experimental platforms that can produce longitudinal measurements of molecular, cellular, and tissue systems as they evolve and generate interventional data across relevant contexts. These data could help world models learn both the natural dynamics of a system and how those dynamics change under intervention. Models will also need to integrate multimodal measurements over long histories to predict possible future states of underlying biological systems. Yet biomedical data will inevitably remain incomplete and heterogeneous. World models will also need to generalize across batch effects, diverging cell lines, laboratory protocols, population differences, and other sources of biological and technical variation.

Another major challenge is evaluation. Benchmarking should compare predictions against ground-truth information that has not leaked into training data. Multi-round experimental campaigns that alternate between *in silico* simulation and planning and experimental testing of model predictions are another important direction. Efforts of this scale will often exceed what any single academic or industry laboratory can undertake, motivating new forms of collaboration among academic laboratories, philanthropic funders, and the pharmaceutical and AI industries for the development, evaluation, and use of world models.

Finally, world models could support applications ranging from target discovery, by simulating the effects of perturbing candidate targets in the tissues where they function, to cell reprogramming through multistep combinations of transcription factors, to treatment planning that accounts for disease progression and prior therapies. They could also provide synthetic data for training AI models, initial representations for specialized models, and closed-loop environments for AI agents. Together with advances in data generation, model training, and evaluation, world models could contribute to more general models of human biology and disease.

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DECLARATION OF INTERESTS

The authors declare no competing interests.

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